Atom-cavity interface for Quantum Network applications

EE6502 Term Paper

Krishna Balaji S
Department of Physics, IIT Madras, Chennai, India

Quantum networks enable new paradigms in quantum computing by linking spatially separated quantum nodes through quantum communcation channels. A key requirement for such networks is the ability to coherently interface stationary qubits at nodes, which store and process quantum information, with flying qubits that mediate communication across the network. Atoms and photons are natural candidates for these roles, due to their long coherent times (for storing and processing information) and fast speeds respectively. However, achieving efficient and coherent atom-photon interaction poses a challenge due to small interaction cross-sections of these qubits. In this report, we will study techniques to enable coherent atom-photon interactions via a cavity, and present an implementation that integrates this with present state-of-the-art neutral atom quantum computing architectures that shows a promising pathway for realizing robust quantum network architectures based on neutral atom quantum technologies.

I. INTRODUCTION TO QUANTUM NETWORKS

Distributed quantum computing (DQC) is a promising approach to QC that uses multiple quantum processors (aka nodes) in a network architecture. It enables several components with ability to perform autonomous quantum computations to exchange information via quantum communication channels, via which more complex computations can be achieved, thus keeping the load per node at a limit that is more feasible with the present-day noisy quantum computers. To achieve this, two kinds of qubits are necessary - those that are local to the node and can stay coherent for longer than the O(computation time), and those that can be used to transmit this information in a communication channel. Candidate components for each are atoms and photons respectively, due to the long coherent times of individually-controlled atoms as well as fast speeds and non-interacting nature of photonic states.

However, coherent exchange of quantum information between these qubits is not staightforward due to a natural mismatch between the interaction cross-sections of the two, which directly affect their interaction probability. Thus, in the following sections, we will first explore techniques inspired by cavity-QED (cQED) that will help us realize deterministic quantum state transfer [1] between the qubits in an isolated environment. We will characterize cooperativity metric C and use this across implementations to extend this idea of cavity-assisted interaction and study its realization in a practical setting of a neutral-atom quantum computer.

II. ENHANCING LIGHT-MATTER COUPLING WITH CAVITY-ASSISTED INTERACTIONS

A. Atom-Photon interaction

We start by considering an atom as a two-level system whose energy separation is given by w_a, and the light as of wavelength λ such that it is in resonance with the atomic transition ($\omega_a \lambda = 2\pi c$. From the quantum mechanical theory of scattering, it can be shown that at resonance, total scattering cross-section of an atom interacting with a photon of wavelength λ is upper bounded by the value given by [2],

$$\sigma_{max} = \frac{3}{2\pi}\lambda^2$$

In a practical application, this photon is part of a laser beam (with an assumed Gaussian width w_0), and hence can interact with atom within an area given by $A \sim \pi(\frac{w_0}{2})^2$. Probability that an atom absorbs a photon is a function of these two cross-sections, given by $P \sim \sigma_{max}/A$, with some constant pre-factors. Intuitively, this suggests that we want to maximize laser intensity (which is a metric for number of photons) within the atomic absorption cross-section, and having a large area A of the beam leads to spreading out of the available photons over the area, thereby reducing the probability of interaction with the atom.

Thus, to ensure deterministic interaction of the atom with any of the photons of the laser, we impose that $\sigma_{\max} \gg A$. In practice, the control of A for the Gaussian beam is diffraction-limited to beam waists of the order of a few μm (which is of the order of the optical wavelength λ). With some rough back-of-the-envelope calculation, we can arrive at an estimate for σ_{max}/A as,

$$A \sim \pi w_0^2, \quad w_0 \sim \mathcal{O}(\lambda) \sim 2\,\mu m \quad \text{(from a typical experiment)},$$

$$A \sim \pi(2\,\mu m)^2 \approx 12 \times 10^{-12}\,\mathrm{m}^2.$$

The atomic absorption cross section on resonance can be estimated as

$$\sigma_{\max} = \frac{3\lambda^2}{2\pi}.$$

For $\lambda \approx 780\,\text{nm}$ (which is Rb87's D2 transition)[3],

$$\sigma_{\max} \approx \frac{3(780 \times 10^{-9}\,\text{m})^2}{2\pi} \approx 3 \times 10^{-13}\,\text{m}^2.$$

This gives us the estimate as

$$\frac{\sigma_{\max}}{A} \approx \frac{3 \times 10^{-13}}{12 \times 10^{-12}} \ll 1.$$

The interaction probability between the atom and photon can in principle be increased if we were to let the beam hit the atom multiple times, thus increasing the effective atomic cross section as $\sigma_{eff} \sim N\sigma_{max}$, where N is the number of times a given photon passes "through" the atom.

B. Interaction in presence of a cavity

A Fabry-Perot Cavity consists a pair of highly reflective mirrors within which photons are confined. One of the mirrors is designed to be partially transparent so as to allow entry of a photon into it. The reflected waves in the cavity undergo interference, because of which the cavity exhibits sharp resonance peaks, called the modes of the cavity. These form the set of frequencies that can be stored in it. (as a standing wave).

When an atom is made to interact with this sytem, photons trapped inside the cavity can be thought of as bouncing back and forth between the mirrors multiple times before escaping (due to the partially transparent mirror), effectively increasing the interaction time between the photon and the atom placed inside the cavity. (An alternative interpretation of the same effect is that an atom placed at the center (Antinode) of the cavity experiences the maxima of electric field amplitude, which thereby increases probability of scattering of the incident electromagnetic wave off the atom.)

The effect of these mirrors can be quantified as follows:

Let the cavity consist of two mirrors whose reflectivities are given by R_1 and R_2. They are placed on either side of the atom along the beam path (defined by its $\vec{k}$) such that a photon can bounce back and forth many times. The cavity is characterized by a quantity called its Finesse $\mathcal{F}$, which is effectively determined by the modes of the cavity and quality of the mirrors.

The finesse $\mathcal{F}$ can be expressed in terms of the reflectivities as, [1]

$$\mathcal{F} = \frac{\pi(R_1R_2)^{1/4}}{1 - \sqrt{R_1R_2}}.$$

It is formally defined as the ratio of Free-spectral range and bandwidth of resonances in the cavity; we see that it directly relates to reflectivities because R_1, R_2 are effectively losses in the system, and in the limit where $R_1, R_2 \to 1$, $\mathcal{F} \to \infty$, which is expected because this

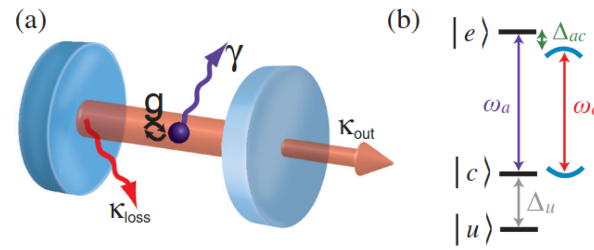

FIG. 1. Atom in a cavity, showing different types of losses that can be incurred

corresponds to a case of perfect resonance and thus the bandwidth $\Delta\omega$ goes to zero. Thus, $\mathcal{F}$ can be used as a metric to quantify the average number of "bounces" that a photon undergoes before it is leaked out of the cavity.

Using our argument from above, we impose the following condition to achieve deterministic interaction atom-photon interaction:

$$\underbrace{\frac{3\lambda^2}{2\pi}}_{\sigma_{\text{abs}}} \times \underbrace{\frac{\mathcal{F}}{\pi}}_{\text{no. of bounces}} \gg \underbrace{\frac{\pi}{4}w_0^2}_{\text{beam area at waist}},$$

where λ is the wavelength of light and w_0 is the beam waist.

This condition can be reformulated in terms of the cavity field decay rate κ, the spontaneous emission rate of the atom γ, and the dipole Rabi coupling constant g, between atom and a cavity mode given by:

$$g = \frac{\mu_{ge}E}{\hbar} = \sqrt{\frac{\mu_{ge}^2\omega}{2\varepsilon_0\hbar V}}\,u(\vec{x}),$$

$$\kappa = \frac{\pi c}{2L\mathcal{F}}, \quad \gamma = \frac{\mu_{ge}^2\omega^3}{6\pi\varepsilon_0\hbar c^3},$$

(references: [1], γ is also the standard Einstein's A coefficient).

We thus define a useful metric called the single-atom Cooperativity C , by writing the above equations purely in terms of the different frequencies in the system:

$$C = \frac{g^2}{\kappa\Gamma} \gg 1$$

Naturally, C is inversely proportional to losses in the system, and depends inversely on the volume of the cavity (i.e. the mode volume V). Larger the C, higher is the probability of photon being absorbed by the atom.

C. Regimes of interaction in a Lossy cavity

Studying dynamics of Atom and Photon in a cavity requires a fully quantized formalism of both Atomic and

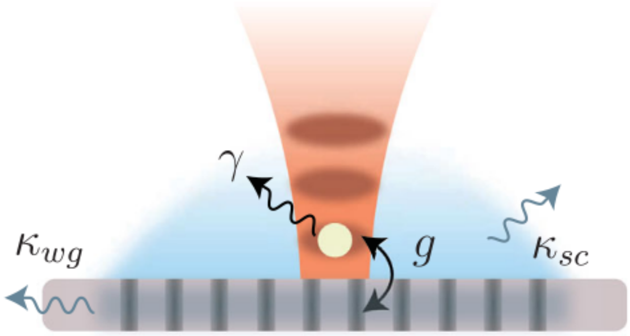

FIG. 2. Neutral atom qubit in a tweezer coupled to a nanophotonic cavity

Photonic states. Their interaction is given by the Jaynes-Cummings model, which has the coupling (interaction) term as follows:

$$H_{int} = \hbar g(\sigma_- a^\dagger + \sigma_+ a)$$

where $\sigma_- = \sigma_+^\dagger$ is the lowering operator for atomic state, and $a, a^\dagger$ are the ladder operators for photonic Fock states. Due to presence of losses, the evolution is non-unitary, and can be solved for by solving the Lindblad master equation.

However, these losses present different regimes of interaction, of which we find a few interesting:

The "strong coupling regime" $C \gg 1, k < g$ enables coherent energy exchange between atom and resonator. However, it is also possible to maintain high Cooperativity ($C > 1$) but still enabling losses such that $k \sim g$ (granted γ is small). Note that k, g can both be tuned in an experiment because they depend on the parameters of Laser, and losses due to the cavity respectively.

D. Purcell regime and atom-photon state transfer

The regime where $C \gg 1, k \lesssim g$ is called "Purcell regime" and shows interesting dynamics. From a quantum network perspective, our final goal is to be able to design an atom-photon interface that allows deterministic state transfer. To enable this, we first try to understand the decay channels in the system.

κ refers to total loss rate *due to cavity*, and can in principle manifest in various forms. There's κ_{sc} which accounts for losses due to scattering off the mirrors, while κ_{out} accounts for leakage of propagating mode from imperfect cavity mirrors. This particular loss is of interest to us because scattering and spontaneous emission losses are random in *direction*, and cannot be used to transmit information at will.

In the Purcell regime, we can enable strong emissions specifically into the **cavity mode** $\vec{k}_{out}$ which is dictated by κ_{out}, as opposed to free-space modes due to spontaneous emission decay rate γ. Thus, in the Purcell regime, by maintaining $\kappa_{out} >> \kappa_{sc}$, one can physically realize a situation where an atom has strong emissions into cavity mode (whose rate is given by $\gamma_c = 2C\gamma$ [1] at $\omega_a = \omega_c$), but since κ_{out} dominates all emission frequencies, the photon escapes the cavity before it can be reabsorbed by the atom. In such a system, when a strong enough excitation pulse is sent we expect to see an emitted photon in the emission mode$\vec{k}_{out}$, thus becoming closer to the notion of deterministic state atom-photon state transfer. While above technique presents the notion of deteministic photon generation, it can be extended to an arbitary initial state by choosing to couple right spin angular momentum m_F states of the atom (which emit different polarizations of light, from conservation of total angular momentum of light-atom system).

III. CAVITY-QED WITH A NEUTRAL-ATOM QUBIT

A. Introducing Nanophotonic cavities

While the previous discussion studies the dynamics of an atom in free-space enclosed in a Fabry-Perot cavity, in a practical setting it is not physical to be able to introduce cavities specific to each atom in a quantum computer having several qubits. A quantum computer made of atom arrays typically has them trapped at distances of μm from each other[4]. Thus, the use of nanophotonic structures as cavities becomes necessary.

Nanophotonic cavities are resonators that allow quantized modes of light at the scale of nanometers. One such example is that of a photonic crystal [5] , which uses spatial modulations of dielectric constant on a waveguide to generate cavity modes that can propagate light. These modulations can be formed from be defects on the surface (Figure 2) that help form discrete optical modes to trap the photons, thus behaving as a cavity.

B. Cooperativity Measurement

The nanophotonic cavities are nanostructures whose total length if of the other of millimeters; thus, they can be smoothly integrated into an experimental setup with the isolated neutral atom qubits. The coupling of the cavity modes is achieved by bringing the crystal in direction transverse to tweezers. Atoms can move along this transverse direction, and can hence be brought in contact with the photonic crystal to enable interaction.

The interactions are enabled by *evanescent* modes of the cavity, since atoms can't be located inside these cavities (otherwise they won't experience the trapping potential from tweezers, and may get kicked off). The atoms are brought arbitrarily close to the surface of the cavity where evanescent field has maximum intensity. By its

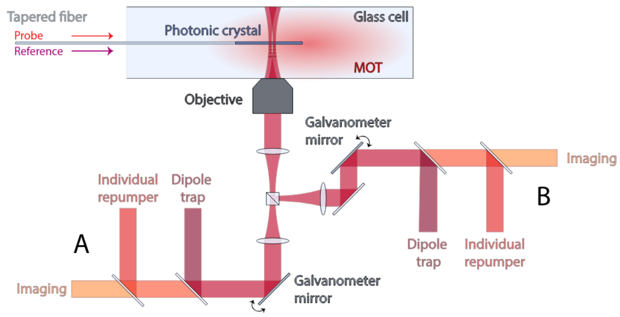

FIG. 3. Experimental setup for integrating a nanophotonic cavity in a Neutral Atom quantum computer

nature, the field decays exponentially radially away from the structure. However, the tweezer light experiences reflections from the structure which causes a standing wave pattern, thus leading to the atom being trapped *better* at places where the standing wave forms an antinode. The nature of their coupling Hamiltonian remains the same, albeit with additional vibrational modes introduced in the atom due to the harmonic trap of the optical tweezer (but these occurs at a much different frequency so it can effectively be ignored). Below we have described the setup as well as discussed the measurement of this coupling using Cooperativity as a metric.

1. *Experimental Setup*

The setup (Figure 3) shows an array of Rb87 atoms individually trapped in an optical tweezer and located [6] inside a (vacuum) glass cell that isolates it from the atmospheric environment. The Photonic crystal is introduced into the cell by attaching it to a tapered optical fiber.

This fiber is used to shine the atom with a Probe beam for the purpose of a reflectivity measurement, which directly allows to determine the system Cooperativity (as per the model presented next).

2. *Cooperativity Model*

Cooperativity C of the system requires the knowledge of the rates g, γ, κ associated with the atom-cavity system. Since the exact values of them are unknown in a general experimental setting, we try to infer it from the reflectivity measurement of the system to a probe frequency. This reflectivity $r(\omega)$ can be modelled as follows:

For an open quantum system (system with losses to the environment), the time evolution of an operator A associated with the system is given by the operator-form of the Lindblad master equation, called the Heisenberg-Langevin relations:

$$\dot{A} = i[H, A] + \sum_i \left(L_i^\dagger A L_i - \frac{1}{2}\left(A L_i^\dagger L_i + L_i^\dagger L_i A \right) \right) \quad (2)$$

where the Hamiltonian H is given by the Jaynes-Cumming interaction,

$$H = \omega_c a^\dagger a + \sum_i \omega_i \sigma_i^\dagger \sigma_i + \sum_i g_i \left(a^\dagger \sigma_i + a \sigma_i^\dagger \right)$$

and operator$A \in \{a, \sigma_i\}$. These equations can be solved in the frequency domain as,

$$\dot{a} = -i\omega_c a - i \sum_j g_j \sigma_j - \frac{\kappa}{2} a + \sqrt{\kappa_{wg}} a_{\text{in}} \quad (1)$$

$$\dot{\sigma}_i = -i\omega_i \sigma_i + ia \sum_j g_j \left(\langle e_j | e_i \rangle - \delta_{ij} \langle g | g \rangle \right) - \frac{\gamma}{2} \sigma_i \quad (2)$$

In the limit of weak excitation (due to low probe intensity), $\langle g|g \rangle \approx 1$, and we can subtitute the below expression for σ_i into the frequency domain a expression:

$$\sigma_i = \frac{g_i a}{\delta_i + i\gamma/2}, \quad \text{where } \delta_i = \omega_i - \omega$$

Substituting into the cavity equation:

$$\left(\frac{\kappa}{2} - i\delta_c \right) a = -i \sum_j g_j \sigma_j + \sqrt{\kappa_{wg}} a_{\text{in}}$$

Thus, we can solve for a as:

$$\frac{a}{a_{\text{in}}} = \frac{\sqrt{\kappa_{wg}}}{\frac{\kappa}{2} - i\delta_c + \sum_j \frac{g_j^2}{\gamma/2 - i\delta_j}}$$

We use the input-output relations between cavity mode operators, given by a_{out}, a_{in} to develop a model for reflectivity r:

$$a_{out} + a_{in} = \sqrt{\kappa_{wg}}\, a$$

$$r = \frac{a_{out}}{a_{in}} = \sqrt{\kappa_{wg}} \frac{a}{a_{in}} - 1$$

Substituting $\frac{a}{a_{in}}$ the above relation into the cavity equation, we obtain the reflectivity spectrum:

$$r = \kappa_{wg} \left(\frac{\kappa}{2} - i\delta_c + \sum_i \frac{g_i^2}{\gamma/2 - i\delta_i} \right)^{-1} - 1$$

Since Cooperativity C directly depends on Rabi coupling with i^{th} mode, g_i, the spontaneous emission rate γ and cavity loss rates κ, κ_{wg}, the unknown parameters

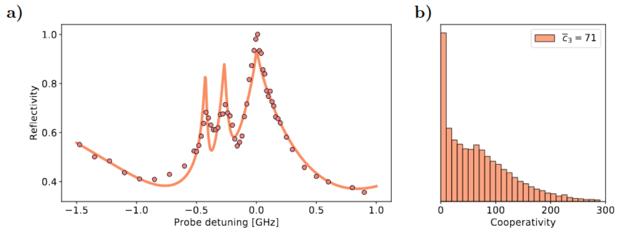

FIG. 4. (a) Fitting reflection model to observed spectrum. The 3 peaks are that of the hyperfine transitions F = 1 to F' = 1,2,3 (spaced ∼ 100MHz apart)
(b) Histogram from multiple rounds of cooperativity estimation.

can be fit into an experimental reflection spectrum r to thus infer Cooperativity of the system as, $C = \frac{g^2}{\kappa\gamma}$. g_i, δ_i are Rabi strengths and detunings of the i^{th} atomic transition from the cavity mode; Since cavity mode is optical and these contributions come mostly from hyperfine splitting of the atomic levels, they can be reasonably approximated to a single value g, δ (with $\delta = 0$ at atomic resonance)

3. Results

For measuring the reflection spectra, a probe beam was sent through the tapered fiber from which atomic emission was captured by a camera.

Results from the Probe spectrscopy are shown in Figure 4. The loss parameters were estimated according to the model described previously as, $2g = 2\pi \times 1.24$ GHz, $\Gamma = 2\pi \times 0.006$ GHz, $\kappa_{\mathrm{wg}} = 2\pi \times 0.86$ GHz, $\kappa_{\mathrm{sc}} = 2\pi \times 2.77$ GHz, and subsequently, cooperativity was calculated as $C = 71 \pm 4$, thus achieving strong coupling of qubit with the evanescent field of photonic crystal, while operating within the Purcell regime.

Spectral lines were observed to be broadened due to "Purcell effect" that was described earlier, which is expected due to an enhanced Spontaneous decay rate γ_c in presence of coupling to the cavity mode.

IV. CONCLUSION

In this report, we have introduced the necessity for atom-cavity interactions from the perspective of applications in a Quantum Network. We proceeded to discuss the realization of deteministic atom-light state transfer in a qubit-qubit interface, for which we took the aid of cQED anaylsis. We introduced Cooperativity as a figure of merit and analyzed the interesting regimes of operation that will enable the aforementioned state transfer between the traveling and flying qubits by considering various losses in the atom-cavity system. We then discussed a practical implementation of the same idea but at the scale of a single qubit in a neutral atom quantum computing architecture, where the role of cavity was realized with a nanophotonic structure. We proceeded to develop a model for Cooperativity via reflectivity measurements, from which the loss parameters of the system were inferred. A high C of 71 ± 4 was achieved, which highlights the importance of this system in facilitating future experiments to test coherent state transfer and thus paves the way to building robust quantum network architectures.

[1] A. Reiserer and G. Rempe, Rev. Mod. Phys. **87**, 1379 (2015).
[2] J. D. Jackson, *Classical Electrodynamics*, 3rd ed. (Wiley, New York, 1998).
[3] D. A. Steck, Rubidium 87 d line data (2021), revision 1.6.
[4] D. Bluvstein, S. J. Evered, A. A. Geim, *et al.*, Nature **626**, 58 (2024).
[5] J. D. Joannopoulos, S. G. Johnson, J. N. Winn, and R. D. Meade, *Photonic Crystals: Molding the Flow of Light*, 2nd ed. (Princeton University Press, Princeton, NJ, 2008).
[6] P. Samutpraphoot, T. orević, P. L. Ocola, H. Bernien, C. Senko, V. Vuletić, and M. D. Lukin, Phys. Rev. Lett. **124**, 063602 (2020).